

第62回 コンピュータビジョン勉強会@関東

Long-CLIP:

Unlocking the Long-Text Capability of CLIP

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自己紹介: 川田拓朗

経歴

- '21/04～ 法政大学 彌富研 B4

研究分野

- Vision & Language, Infographic に興味があります
- 学術論文における Graphical Abstract 自動生成の初期検討 [[Kawada+, YANS'24](#)]

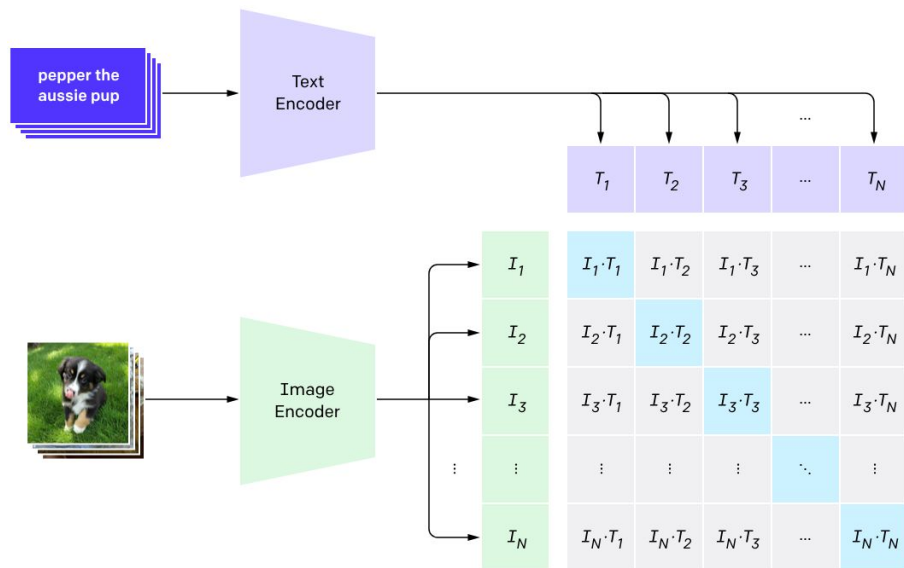


X @lychee1223_Lab

はじめに

CLIP [Radford+, ICML'21]

- Vision & Language の基盤モデル (e.g., LLaVA, Stable Diffusion, FLUX ...)
- 画像とテキストを同一空間に埋め込む



はじめに

CLIPのテキストエンコーダ

- 77トークンの制限
- 絶対位置埋め込みを採用

個人的な所感

- 😊 圧倒的バッチサイズ (32k) で学習された, 視覚と言語の統合的な埋め込み！
- 😊 直感的でリーズナブルな対照学習
- 😞 深い意味理解ができない...
 - 入力テキストが長く, 複雑になるとポンコツになりがち...
 - 細かなニュアンスを汲み取ってくれない...

CLIP、限界かも。

Long-CLIP: Unlocking the Long-Text Capability of CLIP [Zhang+, ECCV'24]

- CLIP のテキスト制限を 77 → 248 トークンに拡張したモデル
- CLIP の潜在空間との整合性を保ち, 様々な下流タスクで置換可能

Image-Text Retrieval

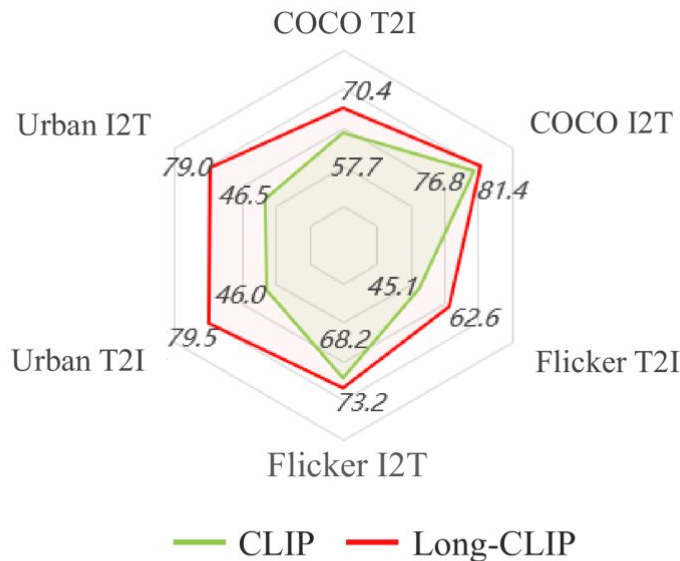
The image shows a city street scene . On the left side, there is a red brick building with a series of **white columns and lamp posts leading down a sidewalk**. A few pedestrians are walking on the sidewalk. There is a **blue mailbox and a black trash can** near the pedestrians. In the center of the image, there is a **multi-lane road**. To the right, there are **more buildings** including a **tall beige building with many windows**, and a shorter red brick building. **Street traffic lights** are visible. The road curves slightly to the right in the distance.



CLIP ✗



Long-CLIP ✓



Long-CLIP: Unlocking the Long-Text Capability of CLIP [Zhang+, ECCV'24]

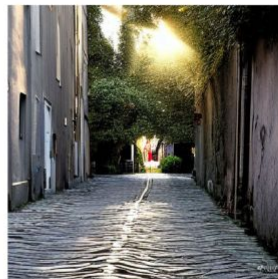
- CLIP のテキスト制限を 77 → 248 トークンに拡張したモデル
- CLIP の潜在空間との整合性を保ち, 様々な下流タスクで置換可能

Text-to-Image Generation

This picture shows a quiet and beautiful park. In distance, there is a **mountain that reaches to the sky**. In the park, a deep path leads to a distant forest. The trees on both sides of the road are **maples**, and **yellow maple leaves are falling on the path**.

The afternoon sun casts diagonal rays onto the alley, creating intertwining shadows on the cobblestone path. An **antique street lamp** stands in a corner, patiently awaiting the arrival of dusk. An **old-fashioned bicycle** is parked against the wall, seemingly waiting for its next rider. The Alley is filled with tranquility, a testament to the stillness of time.

Text



CLIP

Long-CLIP

CLIPの課題分析

1

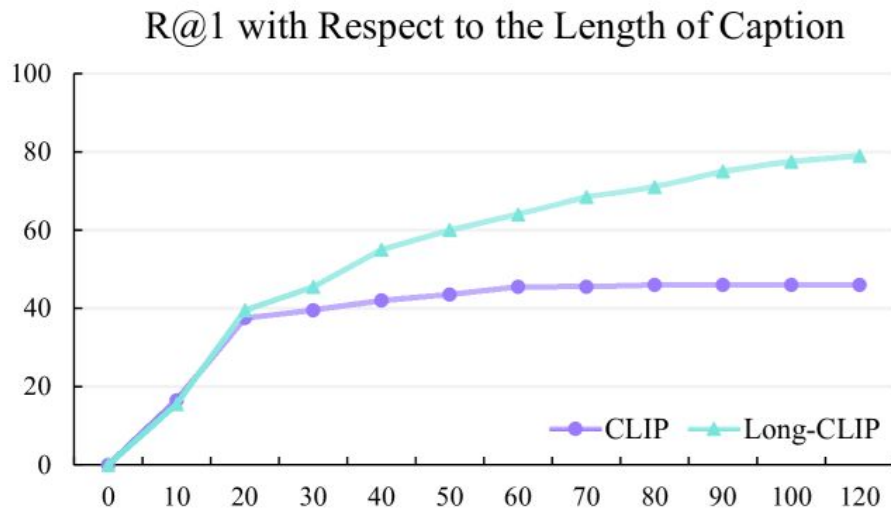
有効なテキストトークンが短い...

2

物体の属性を理解できない...

CLIPの課題分析: 有効なテキストトークンが短い

- Text2Image Retrieval において, query のトークン数と R@1 の関係を調査
→ 🙄 有効なトークン長は20まで



a) Low effective token length

CLIPの課題分析: 物体の属性を理解できない

- Text2Image Retrieval において, query の位置や色を変更し, 類似度を比較
→ 😞 属性が異なっても高い類似度を示す



- A. The lemon on the **left** is **yellow** and the eggplant on the **right** is **purple**
- B. The lemon on the **left** is **purple** and the eggplant on the **right** is **yellow**
- C. The lemon on the **right** is **yellow** and the eggplant on the **left** is **purple**
- D. The lemon on the **right** is **purple** and the eggplant on the **left** is **yellow**

CLIP: $D > C > B > A$ Long-CLIP: $A > C > B > D$

b) Fail in modeling interrelationship

提案手法

1

KPS: Knowledge Preserving Stretching
- 有効な位置埋め込みの拡張法

2

PCM: Primary Component Matching
- 長いテキスト・短いテキストの両方に対応した学習手法

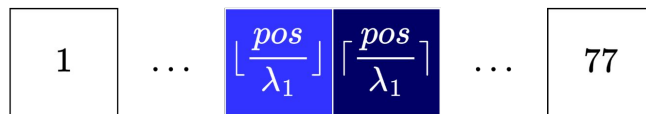
提案手法: KPS (Knowledge Preserving Stretching)

単純に位置埋め込みを線形補完で拡張すると... ?

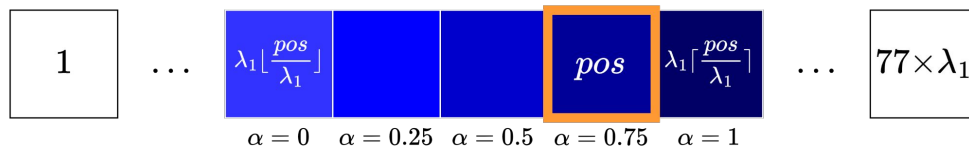
- 😞 引き延ばされて, 事前学習で確立された表現が乱れる
- 😞 有効な20トークン目までを活かせない

$$PE^*(pos) = (1 - \alpha) \times PE(\lfloor \frac{pos}{\lambda_1} \rfloor) + \alpha \times PE(\lceil \frac{pos}{\lambda_1} \rceil), \quad \alpha = \frac{pos \% \lambda_1}{\lambda_1}$$

CLIPの位置埋め込み



λ_1 倍に拡張された位置埋め込み



提案手法: KPS (Knowledge Preserving Stretching)

単純に位置埋め込みを線形補完で拡張すると...?

- 😞 引き延ばされて, 事前学習で確立された表現が乱れる
- 😞 有効な20トークン目までを活かせない

そこで, Long-CLIPでは...

- 20トークン目までは保持し, それ以降は線形補完

$$PE^*(pos) = \begin{cases} PE(pos), & pos \leq 20 \\ (1 - \alpha) \times PE(\lfloor \frac{pos}{\lambda_2} \rfloor) + \alpha \times PE(\lceil \frac{pos}{\lambda_2} \rceil), & \alpha = \frac{pos \% \lambda_2}{\lambda_2}, \text{ otherwise} \end{cases}$$

提案手法: PCM (Primary Component Matching)

単純に長いテキストでFine-tuningすると...?

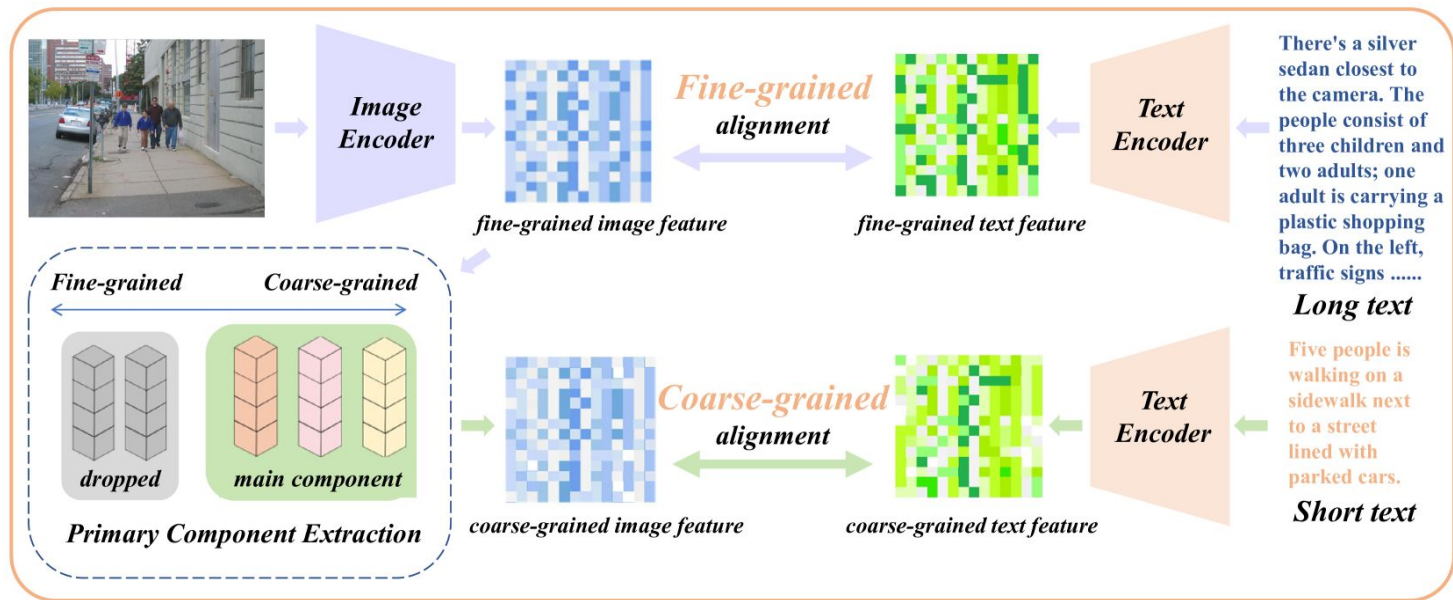
- Text Encoderは長いテキストのみに適応...
- Image Encoderは画像の高周波成分のみを抽出するように...
→ 🙄 短いテキストに対する能力が低下

そこで, Long-CLIPでは..

- 画像とテキストを粒度ごとに分けて算出した対照損失の和を最小化

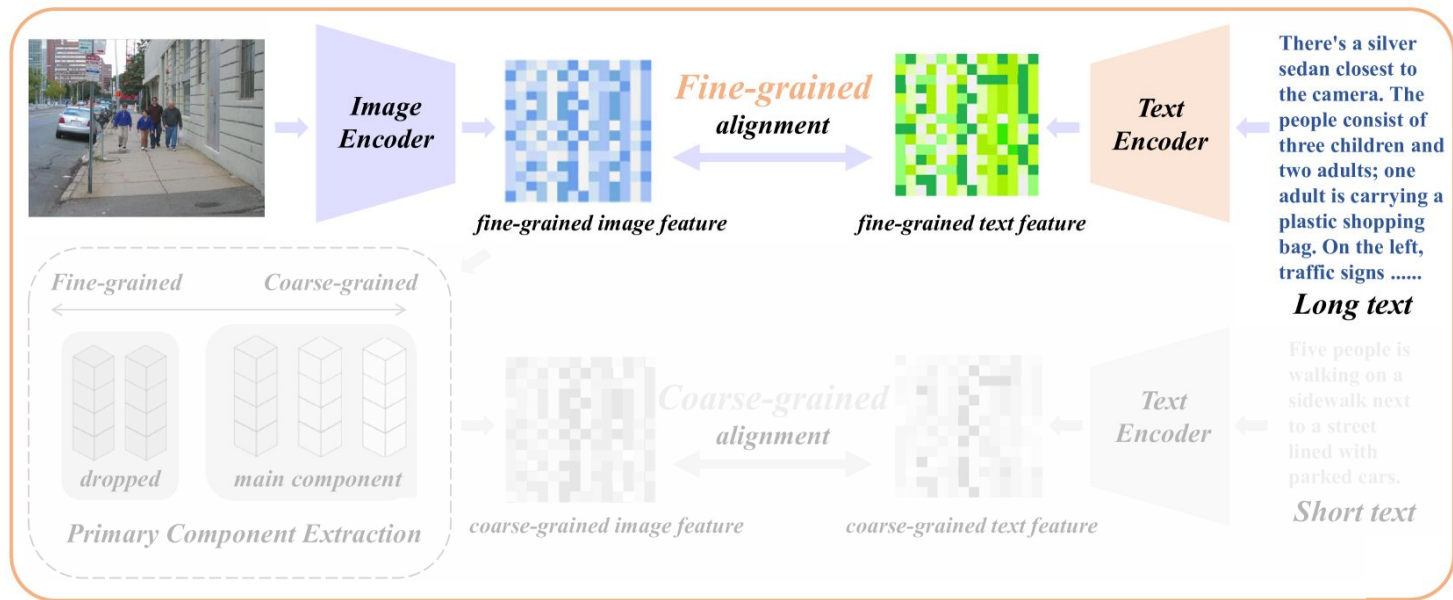
提案手法: PCM (Primary Component Matching)

1. 画像と長いテキストで対照損失を計算
2. 画像の主成分を抽出 (PCAで第32主成分まで)
3. 画像の主成分と短いテキストで対照損失を計算



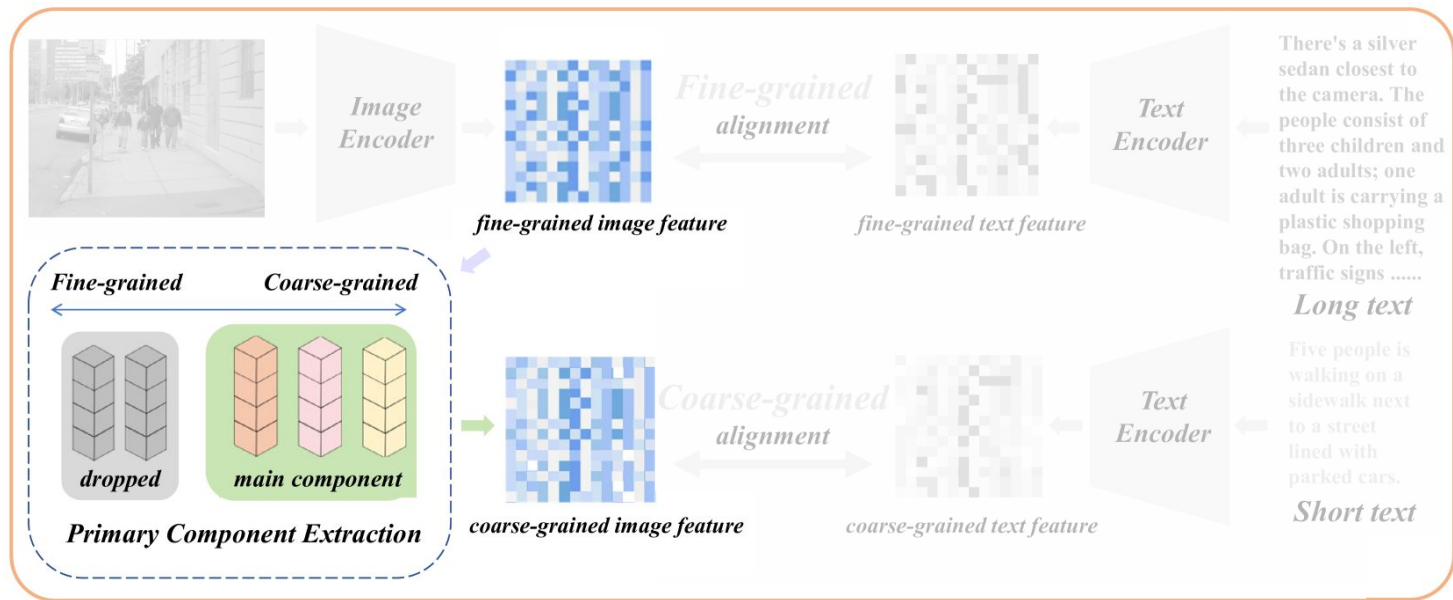
提案手法: PCM (Primary Component Matching)

1. 画像と長いテキストで対照損失を計算
2. 画像の主成分を抽出 (PCAで第32主成分まで)
3. 画像の主成分と短いテキストで対照損失を計算



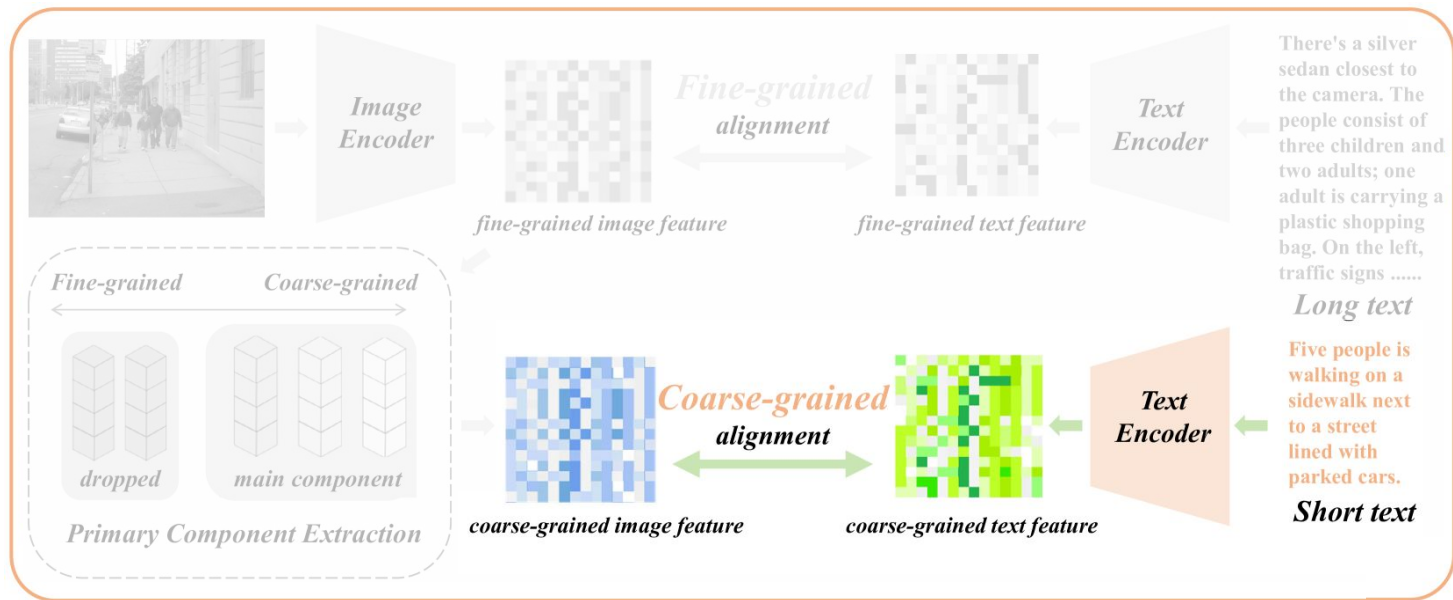
提案手法: PCM (Primary Component Matching)

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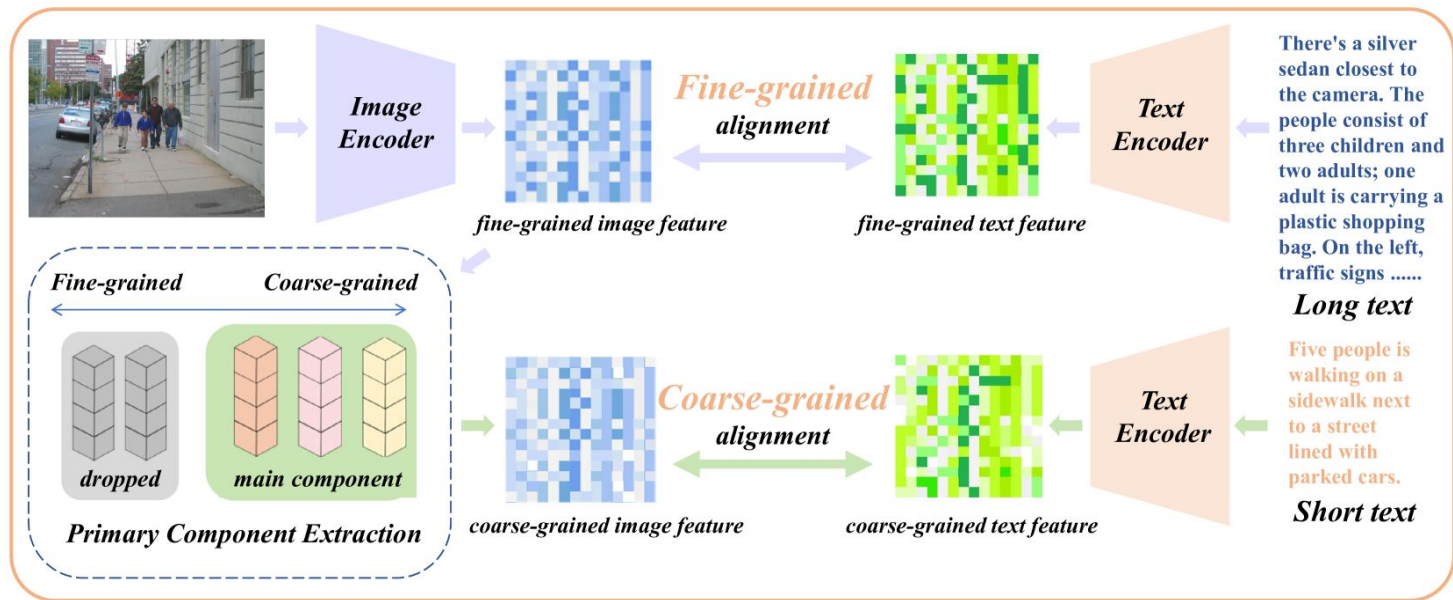
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提案手法: PCM (Primary Component Matching)

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3. 画像の主成分と短いテキストで対照損失を計算



提案データセット: Urban-200

- ベースはVisual genomeデータセット [[Krishana+, CVPR'16](#)]
- 画像とGPT4Vで生成された長いキャプションのペアで構成される
- 提案手法の評価に用いる
- 現在は 200 → 1k に拡張されている



A man in a black jacket is crossing a busy city street. The street is filled with cars of various colors, including **yellow taxis and red trucks**. A traffic light hangs overhead, **currently displaying a green signal**. The perspective of the photo is from the sidewalk, giving a sense of being part of the city's hustle and bustle. The sky above is clear, suggesting good weather. The street is lined with tall buildings and trees, creating a vibrant cityscape.



学習用データセット

ShareGPT4V [Chen+, ECCV'24]

- 画像とテキストのペア100万を学習に使用
- 人手の短いキャプション, GPT4Vで生成された長いキャプションを含む



COCO: Young children standing on a platform waiting for a train to arrive. Adults and children watching a train slowly leave. A family near a railroad track watching the train pass. People waiting on a platform as a train pulls up. A train station with a green train on the tracks and children waiting for it to go by.

LLaVA: At a train station, a group of people, including both young children and adults, are standing on a platform waiting for a train to arrive. The train is already present on the tracks, partially visible on the right side of the image. Some of the people watch the train closely, while others seem to be patiently anticipating its departure.

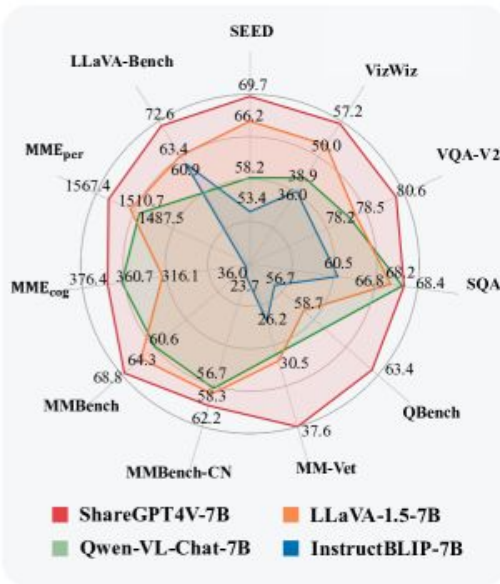
There is a total of eight individuals waiting for the train, with one child in the middle of the platform and the others scattered around. A backpack can be found on the far left side of the platform, suggesting that someone may have set it down while waiting.

ShareGPT4V: The image captures a moment at a train station. A green train is on the tracks, moving away from the platform labeled "Platform 2". The train's motion is observed by people standing on the platform, their attention drawn to the departing vehicle.

A red signal stands on the tracks, its vibrant color contrasting with the green of the train. Signs are prominently displayed around the platform. One warns "Beware of Trains", a cautionary reminder of the powerful machines that frequent these tracks. Another sign advises "Please Use The Footbridge To Cross The Line. If You Require Assistance Please Ask The Station Staff", guiding passengers to safely navigate the station.

The sky above is a clear blue, providing a serene backdrop to the bustling activity of the station. In the distance, trees can be seen, adding a touch of nature to this man-made setting. The image is a snapshot of everyday life at a train station, capturing both its routine operations and its inherent charm.

(a) Comparison of Captions' Quality



(b) Comparison of Performance

評価用データセット

以下のデータセットで様々な下流タスクに対する Long-CLIP の性能を評価

短いテキストと画像のペアからなるデータセット

- COCO2017 [[Lin+, ECCV'14](#)]
- Flickr30k [[Young+, TACL'14](#)]
- ImageNet-1k [[Deng+, CVPR'09](#)]
- ImageNet-V2 [[Recht+, NeurIPS'19](#)]
- ImageNet-O [[Hendrycks+, CVPR'21](#)]
- CIFAR-10 [[Krizhevsky+](#)]
- CIFAR-100 [[Krizhevsky+](#)]

長いテキストと画像のペアからなるデータセット

- ShareGPT4V
- Urban-200

実験: Image-Text Retrieval

1. 長いテキストと画像のペアからなるデータセットを用い, 検索タスクを実施
 - CLIPは77トークン以降は切り捨て
 - R@1で評価

		ShareGPT4V		Urban-200	
		Image-to-Text	Text-to-Image	Image-to-Text	Text-to-Image
B/16	CLIP	78.2	79.6	46.5	46.0
	Direct Fine-tuning	94.1	93.6	78.5	78.0
	Long-CLIP (Ours)	94.6	93.3	79.5	79.0
L/14	CLIP	81.8	84.0	47.0	47.0
	Direct Fine-tuning	95.3	95.4	78.0	76.5
	Long-CLIP (Ours)	95.8	95.6	81.5	81.5

→ 長いテキストにおいて, 大幅な性能向上

実験: Image-Text Retrieval

2. 短いテキストと画像のペアからなるデータセットを用い, 検索タスクを実施
 - R@kで評価

		COCO						Flickr30k					
		Image-to-Text			Text-to-Image			Image-to-Text			Text-to-Image		
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
B/16	CLIP	51.8	76.8	84.3	32.7	57.7	68.2	44.1	68.2	77.0	24.7	45.1	54.6
	Direct Fine-tuning	37.4	62.3	72.1	21.8	43.4	54.5	25.7	45.8	55.4	17.9	34.5	43.1
	Long-CLIP(Ours)	57.6	81.1	87.8	40.4	65.8	75.2	46.8	71.4	79.8	34.1	56.3	65.7
L/14	CLIP	56.1	79.5	86.8	35.4	60.1	70.2	48.5	72.6	80.8	28.0	49.3	58.7
	Direct Fine-tuning	37.9	63.1	72.2	23.1	45.1	55.9	26.0	46.3	55.6	17.9	34.9	43.5
	Long-CLIP(Ours)	62.8	85.1	91.2	46.3	70.8	79.8	53.4	77.5	85.3	41.2	64.1	72.6

→ 短いテキストにおいても性能向上

実験: Image-Text Retrieval

Image-Text Retrieval

The image captures a serene coastal town, nestled on a hillside. The town is characterized by a series of **white houses, each topped with a vibrant red roof**. These houses are scattered across the hillside, **their white facades contrasting beautifully with the red roofs**. The hillside itself is adorned with lush green trees and bushes, adding a touch of nature to the urban landscape. **The town overlooks a deep blue body of water, which is visible in the foreground of the image.** The water's surface is calm, reflecting the clear blue sky above. **A few clouds are scattered across the sky, adding depth to the scene.** The image is taken from a distance, providing a panoramic view of the town and its surroundings. The perspective allows for a comprehensive view of the town, the water, and the sky, creating a harmonious blend of urban life and natural beauty.



CLIP ✗



LongCLIP(Ours) ✓

実験: zero-shot 画像分類

1. 各クラスの代表埋め込みと画像の類似度を測り, Top1のクラス分類
 - ラベル, 設定は CLIP に従う
 - Top1 accuracy で評価

		ImageNet	ImageNet-O	ImageNet-V2	Cifar10	Cifar100	Average
B/16	CLIP	68.4	42.2	61.9	90.8	67.3	66.12
	Direct Fine-tuning	55.1	31.7	44.8	83.9	59.2	54.94
	Long-CLIP(Ours)	66.8	42.7	61.2	90.7	69.3	66.14
L/14	CLIP	75.5	31.9	69.9	95.5	76.8	69.92
	Direct Fine-tuning	58.4	29.2	52.7	92.7	68.7	60.3
	Long-CLIP(Ours)	73.5	33.7	67.9	95.3	78.5	69.78

→ 粗い粒度の特徴を取得する分類タスクにおいても
大幅な性能低下は見られない

実験: 各提案手法の影響調査

1. 長いテキストでのFine-tuningにおいて,
KPS と PCM それぞれの適用がどのような影響を与えるか調査

KPS	PCM	ImageNet	Cifar100	COCO T2I R@5	Flickr I2T R@5	urban T2I R@1
✗	✗	55.1	59.2	43.4	45.8	78.0
✗	✓	58.8	63.5	46.1	46.0	76.5
✓	✗	65.6	65.9	64.3	70.4	78.0
✓	✓	66.8	69.3	65.8	71.4	79.0

- 共に性能向上に寄与
特に, KPS が大きく性能向上に寄与

実験: 長いテキストでの Fine-tuning における, 他の戦略

Undistinguished Image Feature

- 画像特徴の主成分の抽出を行わず, 画像特徴を粒度で分けずに学習

Mixed-length Text

- 長いテキスト : 短いテキスト = 9 : 1 で無作為に混合

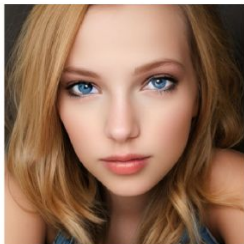
Bounded Text Encoder

- 長いテキストと短いテキストの埋め込みの SmoothL1 をペナルティとして追加

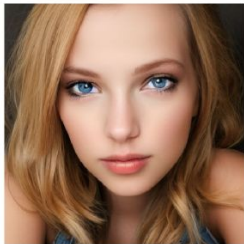
Strategy	ImageNet	Cifar100	COCO T2I R@5	Flickr I2T R@5	Urban-200 T2I R@1
Undistinguished	65.5	67.5	64.6	66.8	71.0
Mixed-length text	66.4	67.8	64.2	68.2	67.5
Bounded encoding	66.8	67.9	65.7	69.3	80.0
Ours	66.8	69.3	65.8	71.4	79.0

実験: SDXL への適用

A beautiful girl with
blonde hair



CLIP



Long-CLIP

The image shows a person standing in front of his farm. He wears a blue hat, **blue T-shirt with red logo** and **black jeans**. He is smiling happily and his **wrinkles is stretching**.



CLIP



Long-CLIP

The early morning sun shines, **dew sparkles** among the green leaves, **birds sing cheerfully**, and the gentle breeze carries the **fragrance of flowers**, creating a vivid picture of vitality.



CLIP



Long-CLIP

The serene lake surface resembled a flawless mirror, reflecting the soft blue sky and the surrounding greenery. A gentle breeze played across its expanse, ruffling the surface into delicate ripples that gradually spread out, disappearing into the distance. Along the shore, weeping willows swayed gracefully in the light breeze, their long branches dipping into the water, creating a soothing sound as they gently brushed against the surface. In the midst of this serene scene, a pure white swan floated gracefully on the lake. Its elegant neck curved into a graceful arc, giving it an air of dignity and elegance that resembled a dancer poised for a performance.



CLIP



Long-CLIP

おわりに

議論

- まだ 248 トークンの上限がある
- 長いキャプションを含むデータセットのさらなる拡張が必要

個人的に使ってみた感想

- CLIPの正当進化！しっかりモデルが強くなってる
- 一方, データの質が重要. その 248 トークンは内容びっしり詰まっていますか？

まとめ

- 😊 テキストの制限を 77 → 248 トークンへ拡張し, CLIPの限界を突破！
- 😊 提案手法 KPS, PCM により, 長文, 短文双方への高性能を実現！
- 😊 幅広い下流タスクにおいて, zero-shot で大幅な性能向上を確認！